

Table I: Microcontroller Series from Some Manufacturers

ATMEL		Microchip Technology			EPSON Semiconductor				Freescale Semiconductor			Infineon		
8 Bit	32 Bit	8 Bit	16 Bit	32 Bit	4 Bit	8 Bit	16 Bit	32 Bit	8 Bit	16 Bit	32 Bit	8 Bit	16 Bit	32 Bit
AT89 Series	UC3 Series	PIC10 and PIC12 Series	PIC24 Series	PIC-32MX Series	SIC6x Family	SIC6x Family	SIC6x Family	SIC6x Family	Freescale S08	Freescale S12	Freescale Kinetis (MQX)	XC800 Family	XE166 Family	Infineon XMC4000
AT90 Series	AT91 SAM (ARM)	PIC16 Series	dsPIC Series						68HC05 (CPU05)	68HC12 (CPU12)	Freescale 683XX		XC 2000 Family	TriCore Family
ATmega		PIC18 Series							68HC08 (CPU08)	68HC16 (CPU16)	MCFSxxx (Freescale Coldfire)		C166 Family	Infineon XMC1000
ATtiny									68HC11 (CPU11)	Freescale DSP56800 (DSP controller)	M-CORE		C167 Family	
ATxmega											MPC500			
											MPC 860 (PowerQUICC)			
											MPC 8240/8250 (PowerQUICC II)			
											MPC 8540/8555/8560 (PowerQUICC III)			

IAR Embedded Workbench, and Open Source tools like the MSPGCC compiler or Energia make it a complete development platform.

STMicroelectronics' Discovery. For students and professionals who are looking to start with ARM architecture, this Cortex-M3 based system would be a great choice. STMicroelectronics actually has come up with several different development boards for virtually every type of ARM microcontroller that it offers. The one shown in Fig. 4 has following features:

1. STM32F051R8T6 microcontroller



Fig. 4: STM32F051R8T6 board

featuring 64kB Flash and 8kB RAM in an LQFP64 package

2. On-board ST-LINK/V2 with selection-mode switch to use the kit as a standalone ST-LINK/V2 (with SWD connector for programming and debugging)

3. Board power supply through USB bus or from an external 5V supply voltage

4. External application power supply: 3V and 5V

5. Four LEDs: LD1 (red) for 3.3V power-on, LD2 (red/green) for USB communication, LD3 (green) for PC9 output, LD4 (blue) for PC8 output

6. Two pushbuttons (user and reset)

7. Extension header for all LQFP64 I/Os for quick connection to prototyping board and easy probing

mbd. The mbd platform provides free software libraries, hardware designs and online tools for professional rapid prototyping of products based on ARM microcontrollers. Hardware of the latest mbed microcontroller boards is based around an NXP microcontroller, which has an ARM Cortex M3 core running at 96 MHz, with 512kB Flash, 64kB RAM, as well as several interfaces including Ethernet, USB

Device, CAN, SPI, I²C and other I/Os.

PandaBoard. PandaBoard is a low-power, low-cost, single-board computer development platform based on Texas Instruments' OMAP4430



Fig. 5: mbed with NXP LPC1768